

software to determine the frequency of available networks. Yet another piece of equipment that the whacker uses for snooping is a GPS system, which is usually connected to the PC and helps map available networks.

So what's the big deal? Well, all these pieces of equipment are mobile. The PC a whacker uses could be a laptop, and car batteries can power the transmitter/receiver setup. Besides, no one will take a second look at a car with an antenna on the roof and a whacker could cruise all around the city listening to everything without anyone noticing him.

But wait, it's not time yet to start looking over your shoulder every time you say those sweet nothings to your significant other over the mobile phone.

Though there is a very real chance that your privacy could be compromised, whacking is more likely to be used for military or industrial espionage than to snoop on individuals.

Although corporate wireless networks are not very common in India, they soon will be. Getting access into wireless LANs is easier because of

the inherent weaknesses in the 802.11 or the Wi-Fi standard that is used for it. Wireless LAN devices make use of the Wired Equivalent Protocol (WEP) for encryption which has several chinks in its armour that make it possible to access it even with off-the-shelf equipment such as wireless LAN cards.

Many of the software that are available for hacking wireless LANs have search engines built into them. These search engines help a snoop find the nearest available wireless access point that he can get at. Some of the software even allows the user to search mobile access points with the name of the service provider as the keyword.

The history of phreaking (breaking into phone lines) speaks of 'war dialling', wherein crackers used to dial up every phone number in the directory to find a number they could crack into. Cracking into wireless LANs have led to the formulation of a term called 'war driving', which involves scouting the streets around target corporate and industrial networks with a PC, which has a wireless LAN card, in search of available access points.

Most wireless LANs can be cracked even without the use of cracking tools. Just think about it—anyone with a PC and a wireless key-

The Ghost in your Mobile

Mobile or cellular phone networks make use of the GSM (Global System for Mobile communications) standard. Most of the communication over the mobile network takes place 'on air'. The first level in signal traffic takes place between the user's handset and the nearest cellular base station. This base station then connects to a core network for the carrier. Security breaches often occur at this point between the base station and the core network. Although there is encryption of signals at the air interfaces between the user and the base station, snoops got over this hurdle (a weak one) a long time back. This was an easy hurdle to cross as there is protection only for the air interface and it is not very strong. The GSM network, taken as a whole, has too many loopholes.

E-commerce over a wireless network too is risky if you are using WAP. In WAP, the core network using a wireless application protocol gateway accesses data on the Internet. The gateway temporarily decrypts the data. Data decryption takes place using WTLS (Wireless Transport Layer Security) which is the security layer in WAP. It is then encrypted back into a form your mobile phone can understand. This encryption is done by SSL (Secure Socket Layer). Now, there is a span of time after the decryption and before the encryption, during which the data is unprotected. At this point of time, it becomes easy to break into the WAP gateway and hackers can forge messages as well as intercept them.

Infographics: Jaya Shetty

